

Evaluation must span tasks, memory, relationships, governance, and time

Layer	Primary question	Evidence	Failure mode	Time scale
Task competence Can it complete work?	success, latency, repair human effort saved	benchmarks, traces counterfactual tests	brittle planning tool misuse	minutes to days
Memory quality Does it remember rightly?	precision, provenance revision, forgetting	memory audits long-horizon QA	false inferences stale profiles	days to years
Relational quality Is adaptation appropriate?	calibration, consent continuity, boundaries	user studies relationship probes	overfamiliarity dependency shaping	weeks to years
Governance Can it be controlled?	permissions, logs explanations	red-team tests policy audits	hidden autonomy unreviewable change	all horizons
Development Does it improve safely?	learning curves drift measures	longitudinal trials ablation suites	identity drift capability regression	months to years